Requirements Collection

To understand what the system needs to do and what users actually expect, two fact-finding techniques were used: **Interviews** and **Research**.

Interview 1 — Traffic Police Officer

Revealed the core hardware problem: handheld devices frequently fail due to extreme heat at desert checkpoints and border areas, forcing officers to fall back on paper-based recording — which is slow, error-prone, and difficult to manage. The officer also highlighted the need for a fast, simple interface that works on any available device.

- Device overheating and battery failure mid-shift
- Paper fallback causes double data entry
- Needs: quick entry, auto-save, any-device access
- Security concern: only authorized officers should record violations

Interview 2 — Government Employee (Traffic Dept.)

Focused on the administrative side. Managing impoundment records manually is time-consuming, and retrieving a specific record requires going through physical files. The employee expressed a clear need for a centralized digital system where records can be searched, updated, and closed quickly.

- Manual record management is slow and error-prone
- Physical file retrieval is inefficient
- Needs: searchable, updatable, closeable digital records

Research — Saudi Arabia's Absher System

Online research focused on the Saudi Arabian traffic violation and management system as a regional reference model. Saudi Arabia operates a fully integrated digital platform called **Absher**, developed by the Ministry of Interior (MOI). Key features studied: role-based access, real-time record updates, online visibility for vehicle owners, digital payment tracking, and automatic SMS notifications within 72 hours of a recorded violation.

- Web-based approach proven at national scale
- Vehicle owners can view violations and pay online
- The gap between Saudi Arabia's system and Jordan's current state confirmed the need

Appendix

Appendix — A Interview

Traffic Police Officer Date: May 2025 | Location: Amman, Jordan

Q1: How do you currently record traffic violations?

We use a handheld device issued by the department. We enter the plate number, violation type, and fine. But the device causes frequent problems, especially in summer on the border areas.

Q2: What kind of problems do you face with the device?

In hot areas near the desert and border checkpoints, the device overheats and shuts down. Sometimes the battery dies mid-shift. When that happens, we write on paper — which means double the work later when someone has to enter it into the system manually.

Q3: How do you feel about using a web-based system instead?

If it works on a phone or laptop, that would be much better in case the dedicated device malfunctions. As long as I can enter the violation quickly and it's saved automatically, that's ideal.

Q4: What information do you need to enter when recording a violation? Plate number, violation type, fine amount, and whether the vehicle needs to be impounded.

Q5: Do you have any security concerns about a digital system?

Yes — only authorized officers should be able to record violations.

Appendix — B Research

#	Website	Purpose
1	**https://www.absher.sa**	Official Saudi MOI Absher portal — main reference
2	**https://syarah.com/carsguide/en/traffic-violations-check/**	Guide to checking violations via Absher
3	**https://lifeinsaudiarabia.net/check-saudi-traffic-violation-fine-details/**	Detailed walkthrough of the Absher violation system
4	**https://www.moi.gov.sa**	Saudi Ministry of Interior — traffic services

Functional Requirements

Functional requirements describe **what the system must do** in order to operate properly. The following are the functional requirements of the Traffic Violation & Impoundment Management System.



User Authentication & Access

- Login via username and password
- Four roles: Admin, Police Officer, Employee, Vehicle Owner
- Role-based access restrictions



Violation Management

- Record violations: plate number, type, fine amount
- Upload vehicle images with violation
- Validate vehicle exists before recording



Impoundment Management

- Create impoundment records and assign to lots
- Authorized employees can release vehicles after conditions are met



Payment & Record Management

- Record payments against violation records
- Close violation records after full payment
- Display remaining balance per record



Search & Visibility

- Search vehicle records by plate number
- Vehicle owners view violations and payment status online
- Admins view all records in the system



Data Management

- Admins manage owners, vehicles, employees, officers
- Admins manage impoundment lots
- All violation records stored permanently in the database

Non-Functional Requirements

Non-functional requirements describe **how the system performs** its functions. The following are the non-functional requirements of the Traffic Violation & Impoundment Management System.

Performance

- Response time under 2 seconds under normal load
- Supports at least 50 concurrent users without degradation

Availability

- 99% uptime — always accessible as hardware backup
- Operational 24/7

Security

- Authentication required before any feature access
- Role-based data access restrictions
- Vehicle owner personal data protected from unauthorized access

Usability

- Simple enough for officers to use with less than 1 hour of training
- Accessible from any modern web browser — no extra software needed

Reliability

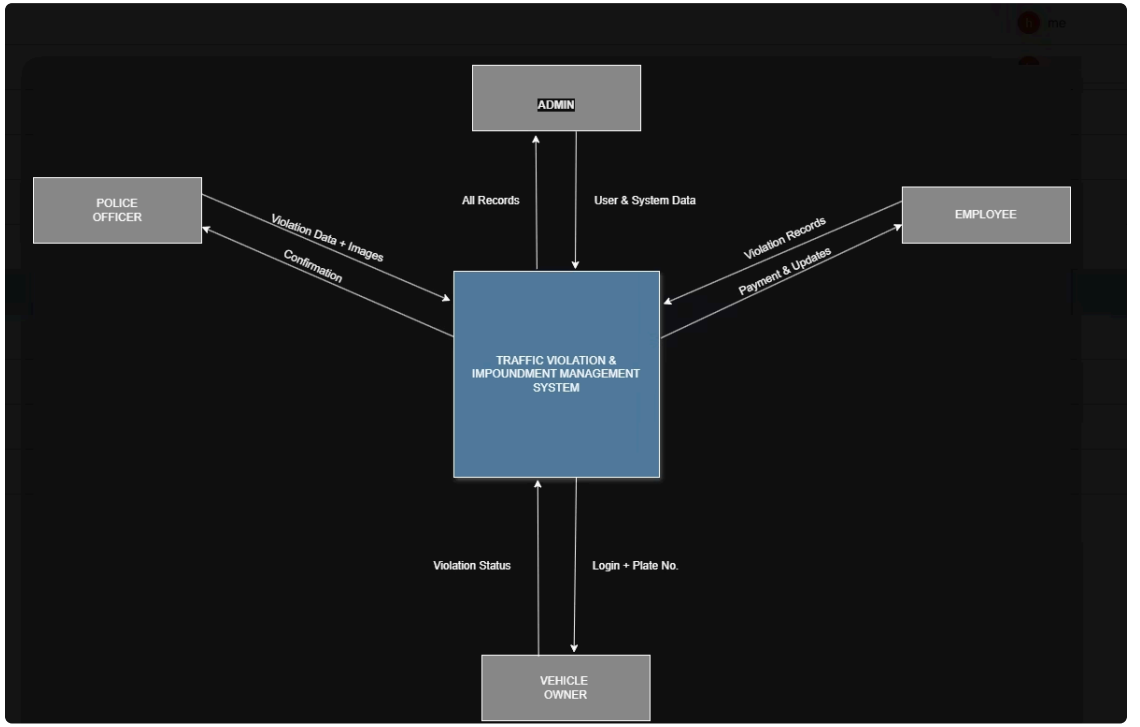
- Auto-save all violation records immediately after entry
- No data loss in case of unexpected shutdown

Scalability & Maintainability

- Designed to migrate from SQLite to MySQL or PostgreSQL as usage grows
- Built with well-documented technologies (Java Spring Boot, SQLite) for easy future maintenance

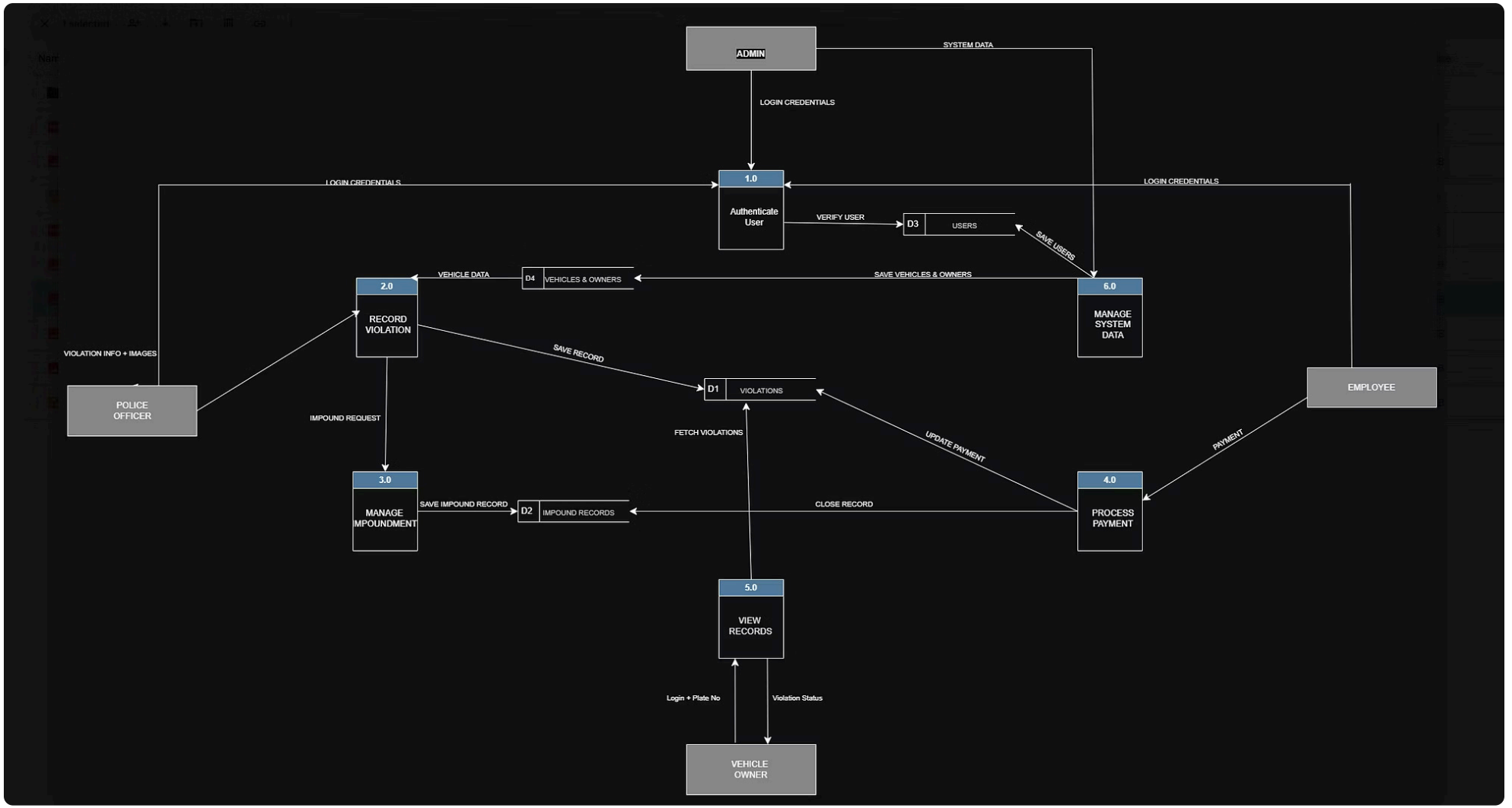
Data Flow Diagram

LEVEL 0



Admin	Police Officer
Employee	Vehicle Owner

LEVEL 1


ADMIN
SYSTEM DATA
LOGIN CREDENTIALS
1.0
Authenticate User
LOGIN CREDENTIALS
LOGIN CREDENTIALS
VERIFY USER
D3
USERS
SAVE USERS
VEHICLE DATA
D4
VEHICLES & OWNERS
SAVE VEHICLES & OWNERS
2.0
RECORD VIOLATION
6.0
MANAGE SYSTEM DATA
VIOLATION INFO + IMAGES
SAVE RECORD
D1
VIOLATIONS
EMPLOYEE
POLICE OFFICER
IMPOUND REQUEST
FETCH VIOLATIONS
UPDATE PAYMENT
PAYMENT
3.0
MANAGE IMPOUNDMENT
SAVE IMPOUND RECORD
D2
IMPOUND RECORDS
CLOSE RECORD
4.0
PROCESS PAYMENT
5.0
VIEW RECORDS
Login + Plate No
Violation Status
VEHICLE OWNER

Process Description Tools - A

Structured English

```
BEGIN Record Violation
  IF officer is authenticated
    IF vehicle plate exists in D4 (Vehicles & Owners)
      Retrieve vehicle and owner details from D4
      Accept violation type, fine amount,
      and images from officer
      IF violation requires impoundment
        Assign vehicle to available impoundment lot
        Send Impound Request to Process 3.0
      ENDIF
      Save violation record to D1 (Violations)
      Return confirmation to Police Officer
    ELSE
      Display error: "Vehicle not found in system"
      Reject violation entry
    ENDIF
  ELSE
    Display error:
    "Unauthorized — officer not authenticated"
    Terminate process
  ENDIF
END Record Violation
```

Process Description Tools - B

Decision Table

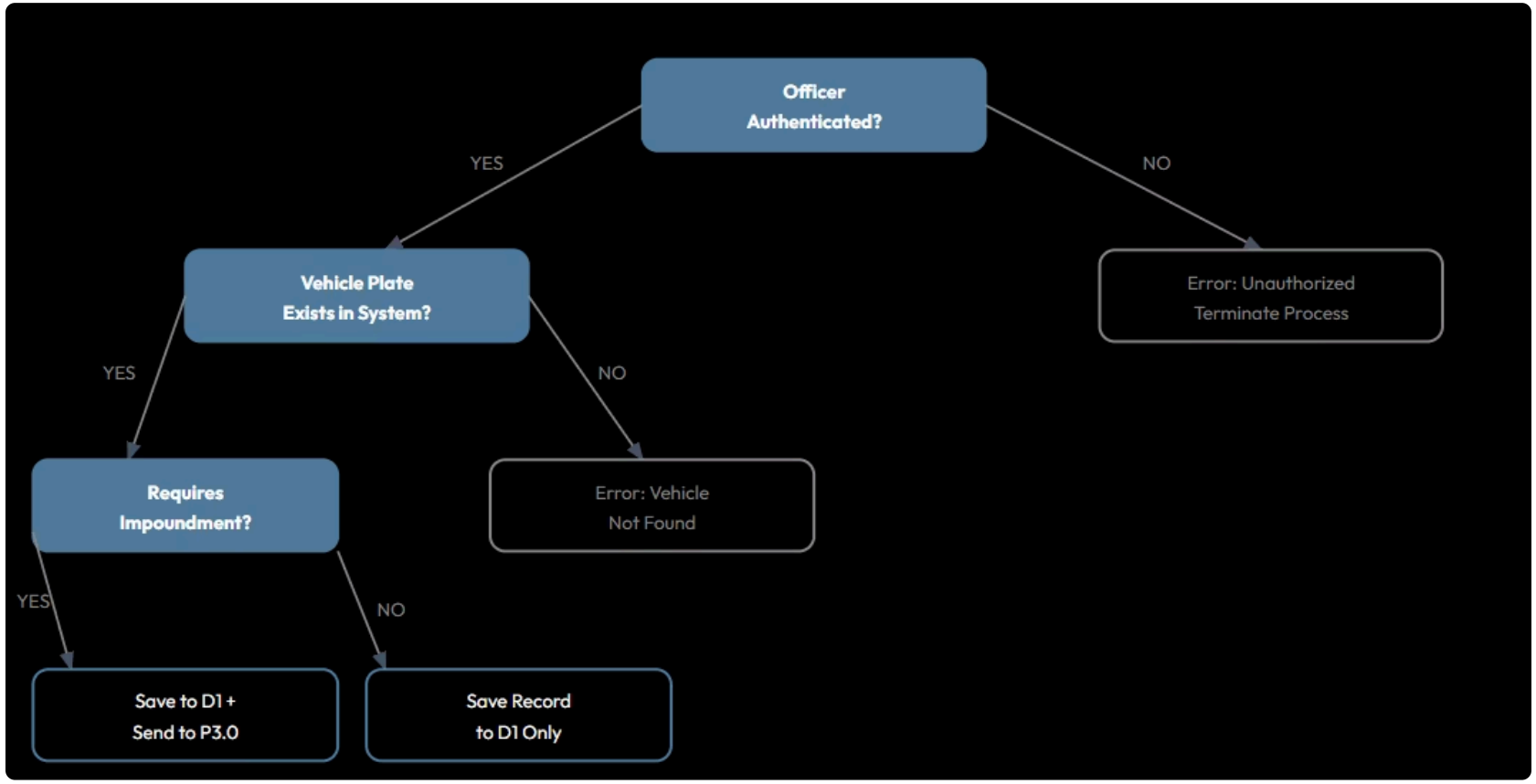
Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
CONDITIONS				
Officer is authenticated?	**Y**	**Y**	**Y**	**N**
Vehicle plate exists in system?	**Y**	**Y**	**N**	–
Violation requires impoundment?	**Y**	**N**	–	–
ACTIONS				
Retrieve vehicle & owner from D4	**X**	**X**	–	–
Accept violation details + images	**X**	**X**	–	–
Assign vehicle to impoundment lot	**X**	–	–	–
Send Impound Request to P3.0	**X**	–	–	–
Save violation record to D1	**X**	**X**	–	–
Return confirmation to officer	**X**	**X**	–	–
Display error: Vehicle not found	–	–	**X**	–
Display error: Unauthorized access	–	–	–	**X**

Condensed Decision Table

Conditions / Actions	Rule 1	Rule 2	Rule 3
CONDITIONS			
Officer authenticated AND vehicle exists?	**Y**	**Y**	**N**
Violation requires impoundment?	**Y**	**N**	–
ACTIONS			
Save to D1 + Send Impound Request to P3.0	**X**	–	–
Save record to D1 only	–	**X**	–
Display error & terminate	–	–	**X**

Process Description Tools - C

Decision Tree



Meeting Minutes

PHASE 2 — ANALYSIS & REQUIREMENTS

Meeting Details

Project: Traffic Violation & Impoundment Management System

Date: May 19, 2025

Time: 7:00 PM

Duration: 1 hour 30 minutes

Location: Online (Google Meet)

Absent Members: None ✅

Next Meeting: May 24, 2025 — Final review before submission

Deadline: May 26, 2025

Attendees & Roles

Name	Role
Juana Hamdan	Requirements Collection & Documentation
Hamza Allan	DFD Design & Process Description
Mohammad Zeyad	Functional & Non-Functional Requirements

Topics Discussed

01

Fact-Finding Techniques

Agreed on Interviews and Research as the two techniques for requirements collection.

02

Primitive Process Selection

Selected P2: Record Violation as the primitive process for structured English, decision table, and decision tree.

03

DFD Work Division

Divided DFD design and process description work among team members.

04

Deadline Review

Reviewed and confirmed submission deadline: May 26, 2025.